

The Cortex

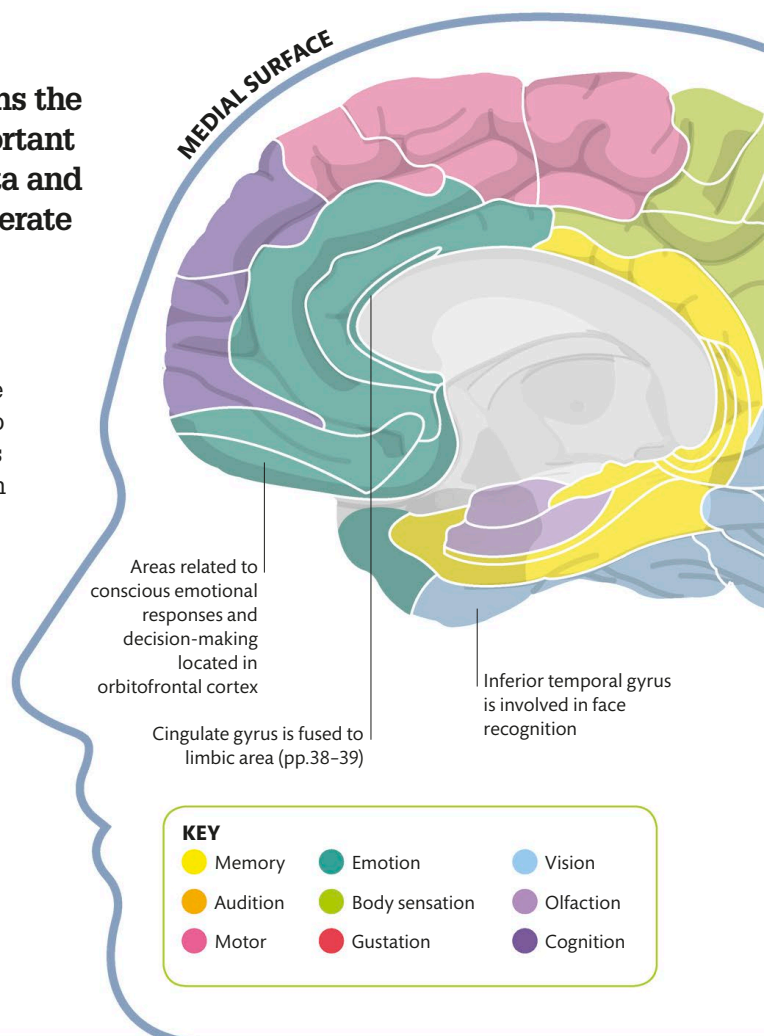
The cortex is the thin outer layer that forms the brain's visible surface. It has several important functions, including handling sensory data and language processing. It also works to generate our conscious experience of the world.

A functional map

The cortex is a multilayered coating of neurons, with their cell bodies at the top. Neuroscientists divide it into areas where the cells appear to work together to perform a particular function. There are different ways to reveal this information: through the location of brain damage linked to the loss of a brain function; tracking the connections between cells; and through scans of live brain activity.

WHAT IS PHRENOLOGY?

A 19th-century pseudoscience, in which the shape of the head was linked to brain structure, specific abilities, and personality.



Folds and grooves

The cerebral cortex is a feature of all mammal brains, but the human brain is distinctive because of its highly folded appearance. The many folds increase the total surface area of the cortex, thereby providing more room for larger cortical areas. The groove in a fold is called a sulcus, and the ridge is called a gyrus. Every human brain has the same pattern of gyri and sulci, which neuroscientists employ to describe specific locations in the cortex.

